

SOLIDWORKS API (.NET VBA C#) — Reference

(279/337): ISketchSpline

SOLIDWORKS API Documentation (converted from CHM)

ISketchSpline (Type)

ISketchSpline Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchSpline Interface

Provides access to sketched spline entities.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISketchSpline
```

Visual Basic (Usage)

```
Dim instance As ISketchSpline
```

C#

```
public interface ISketchSpline
```

C++/CLI

```
public interface class ISketchSpline
```

Visual Basic for Applications (VBA) Syntax

See [SketchSpline](#).

Example

[Set Spline Tangency and Curvature Controls \(VBA\)](#)
[Create Equation-driven Curve \(C#\)](#)
[Create Equation-driven Curve \(VB.NET\)](#)
[Create Equation-driven Curve \(VBA\)](#)

Accessors

[ISketchManager::CreateEquationSpline2](#)

Access Diagram

[SketchSpline](#)

See Also

[ISketchSpline Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISplineHandle Interface](#)
[ISplineParamData Interface](#)

ISketchSpline Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchSpline Interface

The following tables list the members exposed by [ISketchSpline](#).

Public Properties

Name	Description
CurveDegree	Gets or sets the degree of curve for this Bezier curve style spline.

 CurveType	Gets or sets the type of curve for this style spline.
 DisplayControlPolygon	Gets or sets whether to add a control polygon to this spline.
 IsRationalCurve	Gets whether this spline is rational or non-rational.
 IsStyleSpline	Gets whether this spline is a style spline.
 Proportional	Gets or sets whether the spline resizes proportionally when you drag an endpoint the spline.
 ShowCurvatureCombs	Gets or sets whether to show curvature combs.
 ShowInflectionPoints	Gets or sets whether show the inflection points of this spline.
 ShowMinimumRadius	Gets or sets the minimum radius of a curve for this spline.
 ShowSplineHandles	Gets or sets whether to show the handles for this spline.

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Public Methods

Name	Description
 AddCurvatureControl	Adds a curvature control pointer to this spline.
 AddTangencyControl	Adds a new handle to help control the tangency of this spline.
 DeletePoint	Deletes a point on this spline.
 GetControlVertexWeights	Gets the weights of the control vetexes of this rational spline.
 GetEquationParameters	Obsolete. Superseded by ISketchSpline.GetEquationParameters2
 GetEquationParameters2	Gets an equation-driven curve's parameters.
 GetPointCount	Gets the number of points in this sketch spline segment.
 GetPoints	Obsolete. Superseded by ISketchSpline.GetPoints2 and ISketchSpline.IEnumPoints
 GetPoints2	Gets an array of sketch points for the spline.
 GetSplineHandleCount	Gets the number of handles in this spline.
 GetSplineHandles	Gets the handles of this spline.
 IEnumPoints	Gets an enumeration of sketch points for the spline.
 IGetPoints	Obsolete. Superseded by ISketchSpline.GetPoints2 and ISketchSpline.IEnumPoints
 IGetSplineHandles	Gets the handles of this spline.
 InsertPoint	Inserts a point at the specified coordinates of this spline.
 MakeRational	Sets whether this spline is rational or non-rational.
 ModifyControlPoint	Specifies new coordinates for the specified control point of this sketch spline.
 ModifyKnot	Modifies the specified interior knot of this sketch spline.
 RelaxSpline	Smoothens the shape of a spline that was changed by dragging a node on a control polygon .
 ResetAllHandles	Resets all handles to their initial state.
 SetControlVertexWeights	Sets the weights of the control vetexes of this rational spline.
 SetEquationParameters	Obsolete. Superseded by ISketchSpline.SetEquationParameters2
 SetEquationParameters2	Sets an equation-driven curve's parameters.
 Simplify	Reduces the number of points in a spline to increase system performance in models with complex spline curves.

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See Also

[ISketchSpline Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISplineHandle Interface](#)
[ISplineParamData Interface](#)

ISketchSpline Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchSpline Interface

For a list of all members of this type, see [ISketchSpline members](#).

Public Methods

Name	Description
 AddCurvatureControl	Adds a curvature control pointer to this spline.
 AddTangencyControl	Adds a new handle to help control the tangency of this spline.
 DeletePoint	Deletes a point on this spline.
 GetControlVertexWeights	Gets the weights of the control vetexes of this rational spline.
 GetEquationParameters	Obsolete. Superseded by ISketchSpline.GetEquationParameters2
 GetEquationParameters2	Gets an equation-driven curve's parameters.
 GetPointCount	Gets the number of points in this sketch spline segment.
 GetPoints	Obsolete. Superseded by ISketchSpline.GetPoints2 and ISketchSpline.IEnumPoints
 GetPoints2	Gets an array of sketch points for the spline.
 GetSplineHandleCount	Gets the number of handles in this spline.
 GetSplineHandles	Gets the handles of this spline.
 IEnumPoints	Gets an enumeration of sketch points for the spline.
 IGetPoints	Obsolete. Superseded by ISketchSpline.GetPoints2 and ISketchSpline.IEnumPoints
 ISetSplineHandles	Gets the handles of this spline.
 InsertPoint	Inserts a point at the specified coordinates of this spline.
 MakeRational	Sets whether this spline is rational or non-rational.
 ModifyControlPoint	Specifies new coordinates for the specified control point of this sketch spline.
 ModifyKnot	Modifies the specified interior knot of this sketch spline.
 RelaxSpline	Smoothens the shape of a spline that was changed by dragging a node on a control polygon .
 ResetAllHandles	Resets all handles to their initial state.
 SetControlVertexWeights	Sets the weights of the control vetexes of this rational spline.
 SetEquationParameters	Obsolete. Superseded by ISketchSpline.SetEquationParameters2
 SetEquationParameters2	Sets an equation-driven curve's parameters.
 Simplify	Reduces the number of points in a spline to increase system performance in models with complex spline curves.

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See Also

[ISketchSpline Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISplineHandle Interface](#)
[ISplineParamData Interface](#)

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ISketchSpline Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchSpline Interface

For a list of all members of this type, see [ISketchSpline members](#).

Public Properties

Name	Description
 CurveDegree	Gets or sets the degree of curve for this Bezier curve style spline.

 CurveType	Gets or sets the type of curve for this style spline.
 DisplayControlPolygon	Gets or sets whether to add a control polygon to this spline.
 IsRationalCurve	Gets whether this spline is rational or non-rational.
 IsStyleSpline	Gets whether this spline is a style spline.
 Proportional	Gets or sets whether the spline resizes proportionally when you drag an endpoint the spline.
 ShowCurvatureCombs	Gets or sets whether to show curvature combs.
 ShowInflectionPoints	Gets or sets whether show the inflection points of this spline.
 ShowMinimumRadius	Gets or sets the minimum radius of a curve for this spline.
 ShowSplineHandles	Gets or sets whether to show the handles for this spline.

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See Also

[ISketchSpline Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISplineHandle Interface](#)
[ISplineParamData Interface](#)

SOLIDWORKS API Help

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AddCurvatureControl Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : AddCurvatureControl Method (ISketchSpline)

Adds a curvature control pointer to this spline.

.NET Syntax

Visual Basic (Declaration)

```
Function AddCurvatureControl( _
    ByVal XPos As System.Double, _
    ByVal YPos As System.Double, _
    ByVal ZPos As System.Double _
) As SplineHandle
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline
Dim XPos As System.Double
Dim YPos As System.Double
Dim ZPos As System.Double
Dim value As SplineHandle

value = instance.AddCurvatureControl(XPos, YPos, ZPos)
```

C#

```
SplineHandle AddCurvatureControl(
    System.double XPos,
    System.double YPos,
    System.double ZPos
)
```

C++/CLI

```
SplineHandle^ AddCurvatureControl(
    & System.double XPos,
    & System.double YPos,
    & System.double ZPos
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XPos
x coordinate for the pointer
YPos
y coordinate for the pointer
ZPos
z coordinate for the pointer

Return Value

[Spline handle](#)

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::AddCurvatureControl](#).

Example

[Set Spline Tangency and Curvature Controls \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::AddTangencyControl Method](#)
[ISketchSpline::GetSplineHandleCount Method](#)
[ISketchSpline::GetSplineHandles Method](#)
[ISketchSpline::IGetSplineHandles Method](#)
[ISketchSpline::ResetAllHandles Method](#)
[ISketchSpline::ShowCurvatureCombs Property](#)
[ISketchSpline::ShowSplineHandles Property](#)
[ISplineHandle::Curvature Property](#)
[ISplineHandle::CurvatureControlled Property](#)
[ISplineHandle::RadiusOfCurvature Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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AddTangencyControl Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : AddTangencyControl Method (ISketchSpline)

Adds a new handle to help control the tangency of this spline.

.NET Syntax

Visual Basic (Declaration)

```
Function AddTangencyControl( _  
    ByVal XPos As System.Double, _  
    ByVal YPos As System.Double, _  
    ByVal ZPos As System.Double _  
    ) As SplineHandle
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim XPos As System.Double  
Dim YPos As System.Double  
Dim ZPos As System.Double  
Dim value As SplineHandle
```

```
value = instance.AddTangencyControl(XPos, YPos, ZPos)
```

C#

```
SplineHandle AddTangencyControl(  
    System.double XPos,  
    System.double YPos,  
    System.double ZPos  
)
```

```
SplineHandle^ AddTangencyControl(
& System.double XPos,
& System.double YPos,
& System.double ZPos
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XPos
x coordinate of the new handle
YPos
y coordinate of the new handle
ZPos
z coordinate of the new handle

Return Value

[Spline handle](#)

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::AddTangencyControl](#).

Example

[Set Spline Tangency and Curvature Controls \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::AddCurvatureControl Method](#)
[ISketchSpline::GetSplineHandleCount Method](#)
[ISketchSpline::GetSplineHandles Method](#)
[ISketchSpline::ISketchSpline::GetSplineHandles Method](#)
[ISketchSpline::ResetAllHandles Method](#)
[ISketchSpline::ShowSplineHandles Property](#)
[ISplineHandle::TangentDriving Property](#)
[ISplineHandle::TangentMagnitude Property](#)
[ISplineHandle::TangentPolarDirection Property](#)
[ISplineHandle::TangentRadialDirection Property](#)

Availability

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CurveDegree Property (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : CurveDegree Property (ISketchSpline)

Gets or sets the degree of curve for this Bezier curve style spline.

.NET Syntax

Visual Basic (Declaration)

```
Property CurveDegree As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISketchSpline
Dim value As System.Integer

instance.CurveDegree = value

value = instance.CurveDegree
```

C#



Copy Code

```
System.int CurveDegree {get; set;}
```

C++/CLI

```
property System.int CurveDegree {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Degree of curve or -1 if not a Bezier curve style spline

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::CurveDegree](#).

Remarks

This property is only valid for a style spline whose [curve_type](#) is [swStyleSplineCurveType_e](#), BezierCurve.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::IsStyleSpline Property \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

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CurveType Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : CurveType Property (ISketchSpline)

Gets or sets the type of curve for this style spline.

.NET Syntax

Visual Basic (Declaration)

```
Property CurveType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Integer
```

```
instance.CurveType = value
```

```
value = instance.CurveType
```

C#

```
System.int CurveType {get; set;}
```

C++/CLI

```
property System.int CurveType {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of curve as defined in [swStyleSplineCurveType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::CurveType](#).

Example

[Get Style Spline Curve Type \(C#\)](#)
[Get Style Spline Curve Type \(VB.NET\)](#)
[Get Style Spline Curve Type \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::CurveDegree Property \(\)](#)
[ISketchSpline::IsStyleSpline Property \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

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DeletePoint Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : DeletePoint Method (ISketchSpline)

Deletes a point on this spline.

.NET Syntax

Visual Basic (Declaration)

```
Function DeletePoint( _  
    ByVal Point As SketchPoint _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim Point As SketchPoint  
Dim value As System.Boolean  
  
value = instance.DeletePoint(Point)
```

C#

```
System.bool DeletePoint(  
    SketchPoint Point  
)
```

C++/CLI

```
System.bool DeletePoint(  
& SketchPoint^ Point  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point
[Sketch.point](#) to delete (see **Remarks**)

Return Value

True if the point is deleted, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::DeletePoint](#).

Remarks

This method does not work if Point specifies the start or end point of this spline.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetPointCount Method](#)
[ISketchSpline::GetPoints2 Method](#)
[ISketchSpline::InsertPoint Method](#)
[ISketchSpline::!GetPoints Method](#)
[ISketchSpline::!EnumPoints Method](#)
[ISplineHandle::SplinePointNumber Property](#)
[ISketchSpline::ShowInflectionPoints Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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DisplayControlPolygon Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : DisplayControlPolygon Property (ISketchSpline)

Gets or sets whether to add a control polygon to this spline.

.NET Syntax

Visual Basic (Declaration)

Property DisplayControlPolygon As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Boolean
```

```
instance.DisplayControlPolygon = value
```

```
value = instance.DisplayControlPolygon
```

C#

```
System.bool DisplayControlPolygon {get; set;}
```

C++/CLI

```
property System.bool DisplayControlPolygon {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to add a control polygon, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::DisplayControlPolygon](#).

Example

[Get and Set Spline Properties \(VBA\)](#)

Remarks

To add a control polygon, set [ISldWorks::SetUserPreferenceToggle](#) swSketchShowSplineControlPolygon to True.

If the spline is not smooth after dragging a node of the control polygon, use [ISketchSpline::RelaxSpline](#) to smoothen its shape.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetControlVertexWeights Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetControlVertexWeights Method (ISketchSpline)

Gets the weights of the control vetexes of this rational spline.

.NET Syntax

Visual Basic (Declaration)

```
Function GetControlVertexWeights() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Object
```

```
value = instance.GetControlVertexWeights()
```

C#

```
System.object GetControlVertexWeights()
```

C++/CLI

```
System.Object^ GetControlVertexWeights();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of control vertex weights

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetControlVertexWeights](#).

Example

[Edit Spline \(VBA\)](#)

Remarks

This method:

- is valid only if [ISketchSpline::IsRationalCurve](#) is true.
- returns an empty array if the spline is non-rational.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

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GetEquationParameters Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetEquationParameters Method (ISketchSpline)

Obsolete. Superseded by [ISketchSpline::GetEquationParameters2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetEquationParameters( _  
    ByRef YExpression As System.String, _  
    ByRef RangeStart As System.Double, _  
    ByRef RangeEnd As System.Double, _  
    ByRef IsAngleRange As System.Boolean, _  
    ByRef RotationAngle As System.Double, _  
    ByRef XOffset As System.Double, _  
    ByRef YOffset As System.Double, _  
    ByRef LockStart As System.Boolean, _  
    ByRef LockEnd As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim YExpression As System.String  
Dim RangeStart As System.Double  
Dim RangeEnd As System.Double  
Dim IsAngleRange As System.Boolean  
Dim RotationAngle As System.Double  
Dim XOffset As System.Double  
Dim YOffset As System.Double  
Dim LockStart As System.Boolean  
Dim LockEnd As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.GetEquationParameters(YExpression, RangeStart, RangeEnd, IsAngleRange,  
RotationAngle, XOffset, YOffset, LockStart, LockEnd)
```

C#

```
System.bool GetEquationParameters(  
    out System.string YExpression,  
    out System.double RangeStart,  
    out System.double RangeEnd,  
    out System.bool IsAngleRange,  
    out System.double RotationAngle,  
    out System.double XOffset,  
    out System.double YOffset,  
    out System.bool LockStart,  
    out System.bool LockEnd  
)
```

C++/CLI

```
System.bool GetEquationParameters(  
& [Out] System.String^ YExpression,  
& [Out] System.double RangeStart,  
& [Out] System.double RangeEnd,  
& [Out] System.bool IsAngleRange,  
& [Out] System.double RotationAngle,  
& [Out] System.double XOffset,
```

```
& [Out] System.double YOffset,  
& [Out] System.bool LockStart,  
& [Out] System.bool LockEnd  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

YExpression

Equation for y (see **Remarks**)

RangeStart

Start value of x (see **Remarks**)

RangeEnd

End value of x (see **Remarks**)

IsAngleRange

True if the range and x value represent an angle (in radians), false if not

RotationAngle

Rotation angle (in radians) for the curve

XOffset

Offset in x for f(x), where x = 0 (see **Remarks**)

YOffset

Offset in y for f(x), where x = 0 (see **Remarks**)

LockStart

True to lock the start point (RangeStart) of the curve, false to not

LockEnd

True to lock the end point (RangeEnd) of the curve, false to not

Return Value

True if the equation-driven curve's parameters are returned, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetEquationParameters](#).

Example

[Create Equation-driven Curve \(C#\)](#)

[Create Equation-driven Curve \(VB.NET\)](#)

[Create Equation-driven Curve \(VBA\)](#)

Remarks

YExpression:

Any function that you can define as a 2D function in the following form:

y-in-the-sketch = f(x-in-the-sketch)

will appear as a curve in the sketch. For example:

y = sin(x)

RangeStart and RangeEnd

You cannot specify string values for RangeStart and RangeEnd; you must specify double values.

XOffset:

In the equation:

y = sin(x) [x=0 to 2Pi]

the start point (0,0) is moved by XOffset in the x direction.

YOffset:

In the equation:

y = sin(x) [x=0 to 2Pi]

the start point (0,0) is moved by YOffset in the y direction.

See Also

[ISketchSpline Interface](#)

[ISketchSpline Members](#)

[ISketchSpline::SetEquationParameters Method](#)

[ISketchManager::CreateEquationSpline Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEquationParameters2 Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetEquationParameters2 Method (ISketchSpline)

Gets an equation-driven curve's parameters.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEquationParameters2( _  
    ByRef XExpression As System.String, _  
    ByRef YExpression As System.String, _  
    ByRef ZExpression As System.String, _  
    ByRef RangeStart As System.Double, _  
    ByRef RangeEnd As System.Double, _  
    ByRef IsAngleRange As System.Boolean, _  
    ByRef RotationAngle As System.Double, _  
    ByRef XOffset As System.Double, _  
    ByRef YOffset As System.Double, _  
    ByRef LockStart As System.Boolean, _  
    ByRef LockEnd As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim XExpression As System.String  
Dim YExpression As System.String  
Dim ZExpression As System.String  
Dim RangeStart As System.Double  
Dim RangeEnd As System.Double  
Dim IsAngleRange As System.Boolean  
Dim RotationAngle As System.Double  
Dim XOffset As System.Double  
Dim YOffset As System.Double  
Dim LockStart As System.Boolean  
Dim LockEnd As System.Boolean  
Dim value As System.Boolean
```

```
value = instance.GetEquationParameters2(XExpression, YExpression, ZExpression, RangeStart,  
RangeEnd, IsAngleRange, RotationAngle, XOffset, YOffset, LockStart, LockEnd)
```

C#

```
System.bool GetEquationParameters2(  
    out System.string XExpression,  
    out System.string YExpression,  
    out System.string ZExpression,  
    out System.double RangeStart,  
    out System.double RangeEnd,  
    out System.bool IsAngleRange,  
    out System.double RotationAngle,  
    out System.double XOffset,  
    out System.double YOffset,  
    out System.bool LockStart,  
    out System.bool LockEnd  
)
```

C++/CLI

```
System.bool GetEquationParameters2(  
& [Out] System.String^ XExpression,
```

```

& [Out] System.String^ YExpression,
& [Out] System.String^ ZExpression,
& [Out] System.double RangeStart,
& [Out] System.double RangeEnd,
& [Out] System.bool IsAngleRange,
& [Out] System.double RotationAngle,
& [Out] System.double XOffset,
& [Out] System.double YOffset,
& [Out] System.bool LockStart,
& [Out] System.bool LockEnd
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XExpression
Equation for x
YExpression
Equation for y
ZExpression
Equation for z
RangeStart
Start value of x
RangeEnd
Start value of y
IsAngleRange
True if the range and x value represent an angle (in radians), false if not
RotationAngle
Rotation angle (in radians) for the curve
XOffset
Offset in x for f(x), where x = 0
YOffset
Offset in y for f(x), where x = 0
LockStart
True to lock the start point (RangeStart) of the curve, false to not
LockEnd
True to lock the end point (RangeEnd) of the curve, false to not

Return Value

True if the equation-driven curve's parameters are returned, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetEquationParameters2](#).

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[SketchManager::CreateEquationSpline2 Method](#)
[ISketchSpline::SetEquationParameters2 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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GetPointCount Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetPointCount Method (ISketchSpline)

Gets the number of points in this sketch spline segment.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPointCount() As System.Integer
```

Visual Basic (Usage)

```

Dim instance As ISketchSpline
Dim value As System.Integer

value = instance.GetPointCount()

```



C#

```
System.int GetPointCount()
```

C++/CLI

```
System.int GetPointCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of points in this sketch spline segment

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetPointCount](#).

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::DeletePoint Method](#)
[ISketchSpline::GetPoints2 Method](#)
[ISketchSpline::IEnumPoints Method](#)
[ISketchSpline::InsertPoint Method](#)
[ISketchSpline::SplinePointNumber Property](#)
[ISketchSpline::ShowInflectionPoints Property](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

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GetPoints Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetPoints Method (ISketchSpline)

Obsolete. Superseded by [ISketchSpline::GetPoints2](#) and [ISketchSpline::IEnumPoints](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetPoints() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Object  
  
value = instance.GetPoints()
```

C#

```
System.object GetPoints()
```

C++/CLI

```
System.Object^ GetPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetPoints](#).

See Also

GetPoints2 Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetPoints2 Method (ISketchSpline)

Gets an array of sketch points for the spline.

.NET Syntax

Visual Basic (Declaration)

```
Function GetPoints2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Object
```

```
value = instance.GetPoints2()
```

C#

```
System.object GetPoints2()
```

C++/CLI

```
System.Object^ GetPoints2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [sketch points](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetPoints2](#).

Example

[Get All Elements of Sketch \(VBA\)](#)
[Simplify Spline \(VBA\)](#)
[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::DeletePoint Method](#)
[ISketchSpline::GetPointCount Method](#)
[ISketchSpline::IEnumerablePoints Method](#)
[ISketchSpline::InsertPoint Method](#)
[ISplineHandle::SplinePointNumber Property](#)
[ISketchSpline::ShowInflectionPoints Property](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

GetSplineHandleCount Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetSplineHandleCount Method (ISketchSpline)

Gets the number of [handles](#) in this spline.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineHandleCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Integer  
  
value = instance.GetSplineHandleCount()
```

C#

```
System.int GetSplineHandleCount()
```

C++/CLI

```
System.int GetSplineHandleCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of handles

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetSplineHandleCount](#).

Remarks

Call this method before calling [ISketchSpline::IGetSplineHandles](#) to get the size of the array for that method.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetSplineHandles Method](#)
[ISketchSpline::ResetAllHandles Method](#)
[ISketchSpline::ShowSplineHandles Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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GetSplineHandles Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : GetSplineHandles Method (ISketchSpline)

Gets the handles of this spline.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSplineHandles() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Object  
  
value = instance.GetSplineHandles()
```

C#

```
System.Object GetSplineHandles()
```

C++/CLI

```
System.Object^ GetSplineHandles();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [spline handles](#)

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::GetSplineHandles](#).

Example

[Get and Set Spline Handles \(VBA\)](#)
[Get and Set Spline Handles \(VB.NET\)](#)
[Get and Set Spline Handles \(C#\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetSplineHandleCount Method](#)
[ISketchSpline::IGetSplineHandles Method](#)
[ISketchSpline::ResetAllHandles Method](#)
[ISketchSpline::ShowSplineHandles Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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IEnumPoints Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : IEnumPoints Method (ISketchSpline)

Gets an enumeration of sketch points for the spline.

.NET Syntax

Visual Basic (Declaration)

```
Function IEnumPoints() As EnumSketchPoints
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As EnumSketchPoints
```

```
value = instance.IEnumPoints()
```

C#

```
EnumSketchPoints IEnumPoints()
```

C++/CLI

```
EnumSketchPoints^ IEnumPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

- in-process, unmanaged C++: Pointer to an array of [enumeration of sketch points](#) or NULL if the operation fails
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetPoints2 Method](#)
[ISketchSpline::InsertPoint Method](#)
[ISketchSpline::DeletePoint Method](#)
[ISketchSpline::GetPointCount Method](#)
[ISplineHandle::SplinePointNumber Property](#)
[ISketchSpline::ShowInflectionPoints Property](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

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IGetPoints Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : IGetPoints Method (ISketchSpline)

Obsolete. Superseded by [ISketchSpline::GetPoints2](#) and [ISketchSpline::IEnumPoints](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPoints() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Double  
  
value = instance.IGetPoints()
```

C#

```
System.double IGetPoints()
```

C++/CLI

```
System.double IGetPoints();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::IGetPoints](#).

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)

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[Send comments](#) on this topic.

IGetSplineHandles Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : IGetSplineHandles Method (ISketchSpline)

Gets the handles of this spline.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSplineHandles( _  
    ByVal Count As System.Integer _  
) As SplineHandle
```

```
Dim instance As ISketchSpline
Dim Count As System.Integer
Dim value As SplineHandle

value = instance.IGetSplineHandles(Count)
```

C#

```
SplineHandle IGetSplineHandles(
    System.int Count
)
```

C++/CLI

```
SplineHandle^ IGetSplineHandles(
    & System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of spline handles

Return Value

- In-process, unmanaged C++: Pointer to an array of [spline handles](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISketchSpline::GetSplineHandleCount](#) to get the value for Count.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetSplineHandles Method](#)
[ISketchSpline::ResetAllHandles Method](#)
[ISketchSpline::ShowSplineHandles Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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InsertPoint Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : InsertPoint Method (ISketchSpline)

Inserts a point at the specified coordinates of this spline.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertPoint( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As System.Boolean
```

Visual Basic (Usage)

Copy Code

```
Dim instance As ISketchSpline
Dim X As System.Double
Dim Y As System.Double
```

```
Dim Z As System.Double
Dim value As System.Boolean

value = instance.InsertPoint(X, Y, Z)
```

C#

```
System.bool InsertPoint(
    System.double X,
    System.double Y,
    System.double Z
)
```

C++/CLI

```
System.bool InsertPoint(
    & System.double X,
    & System.double Y,
    & System.double Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate on this spline
Y
Y coordinate on this spline
Z
Z coordinate on this spline

Return Value

True if a point is added to the spline, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::InsertPoint](#).

Remarks

This method works only if X, Y, and Z specify a location on this spline. You cannot add a new point that is not on this spline.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::DeletePoint Method](#)
[ISketchSpline::GetPointCount Method](#)
[ISketchSpline::GetPoints2 Method](#)
[ISketchSpline::EnumPoints Method](#)
[ISplineHandle::SplinePointNumber Property](#)
[ISketchSpline::ShowInflectionPoints Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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IsRationalCurve Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : IsRationalCurve Property (ISketchSpline)

Gets whether this spline is rational or non-rational.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsRationalCurve As System.Boolean
```

```
Dim instance As ISketchSpline
Dim value As System.Boolean

value = instance.IsRationalCurve
```

C#

```
System.bool IsRationalCurve {get;}
```

C++/CLI

```
property System.bool IsRationalCurve {
    System.bool get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if rational, false if non-rational

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::IsRationalCurve](#).

Example

[Edit Spline \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::MakeRational Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

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IsStyleSpline Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : IsStyleSpline Property (ISketchSpline)

Gets whether this spline is a style spline.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsStyleSpline As System.Boolean
```

Visual Basic (Usage)

Copy Code

```
Dim instance As ISketchSpline
Dim value As System.Boolean

value = instance.IsStyleSpline
```

C#

```
System.bool IsStyleSpline {get;}
```

C++/CLI

```
property System.bool IsStyleSpline {
    System.bool get();
}
```

```
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if a style spline, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::IsStyleSpline](#).

Example

[Get Style Spline Curve Type \(C#\)](#)
[Get Style Spline Curve Type \(VB.NET\)](#)
[Get Style Spline Curve Type \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::CurveDegree Property \(\)](#)
[ISketchSpline::CurveType Property \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

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MakeRational Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : MakeRational Method (ISketchSpline)

Sets whether this spline is rational or non-rational.

.NET Syntax

Visual Basic (Declaration)

```
Function MakeRational(  
    ByVal Val As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim Val As System.Boolean  
Dim value As System.Boolean  
  
value = instance.MakeRational(Val)
```

C#

```
System.bool MakeRational(  
    System.bool Val  
)
```

C++/CLI

```
System.bool MakeRational(  
&    System.bool Val  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val
True to make rational, false to make non-rational

Return Value

True if successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::MakeRational](#).

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::IsRationalCurve Property \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

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ModifyControlPoint Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : ModifyControlPoint Method (ISketchSpline)

Specifies new coordinates for the specified control point of this sketch spline.

.NET Syntax

Visual Basic (Declaration)

```
Function ModifyControlPoint( _  
    ByVal Index As System.Integer, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISketchSpline  
Dim Index As System.Integer  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean  
  
value = instance.ModifyControlPoint(Index, X, Y, Z)
```

C#

```
System.bool ModifyControlPoint(  
    System.int Index,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool ModifyControlPoint(  
& System.int Index,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
1-based index of the control point to modify
X

X coordinate
Y
Y coordinate
Z
Z coordinate

Return Value

True if control point successfully modified, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::ModifyControlPoint](#).

Example

[Edit Spline \(VBA\)](#)

Remarks

Before calling this method, call [ISplineParamData::GetControlPoints](#) and [ISplineParamData::ControlPointsCount](#) to help specify Index and the new coordinates.

After calling this method, you must call [IModelDoc2::ForceRebuild3](#) to update the sketch.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::ModifyKnot Method \(\)](#)
[ISplineParamData::SetControlPoints Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

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ModifyKnot Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : ModifyKnot Method (ISketchSpline)

Modifies the specified interior knot of this sketch spline.

.NET Syntax

Visual Basic (Declaration)

```
Function ModifyKnot( _  
    ByVal Index As System.Integer, _  
    ByVal DKnot As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As ISketchSpline  
Dim Index As System.Integer  
Dim DKnot As System.Double  
Dim value As System.Boolean
```

```
value = instance.ModifyKnot(Index, DKnot)
```

C#

```
System.bool ModifyKnot(  
    System.int Index,  
    System.double DKnot  
    )
```

C++/CLI



```
System.bool ModifyKnot(  
& System.int Index,  
& System.double DKnot  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
1-based index of the interior knot to modify (see **Remarks**)
DKnot
Knot value

Return Value

True if knot successfully modified, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::ModifyKnot](#).

Example

[Edit Spline \(VBA\)](#)

Remarks

Interior knots occur after the first set of 0s and before the last set of 1s in the knot array. If the knot array is [0 0 0 0 0.279240779943874 0.55 0.720759220056126 1 1 1 1], then the interior knots are 0.279240779943874, 0.55, and 0.720759220056126.

Before calling this method, call [ISplineParamData::GetKnotPoints](#) and [ISplineParamData::KnotPointsCount](#) to help specify Index and the new knot value.

After calling this method, you must call [IModelDoc2::ForceRebuild3](#) to update the sketch.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::ModifyControlPoint Method \(\)](#)
[ISplineParamData::SetKnotPoints Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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Proportional Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : Proportional Property (ISketchSpline)

Gets or sets whether the spline resizes proportionally when you drag an endpoint the spline.

.NET Syntax

Visual Basic (Declaration)

Property Proportional As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Boolean
```

```
instance.Proportional = value
```

```
value = instance.Proportional
```

C#

```
System.bool Proportional {get; set;}
```

C++/CLI

```
property System.bool Proportional {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to resize the spline proportionally, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::Proportional](#).

Example

[Get and Set Spline Properties \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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RelaxSpline Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : RelaxSpline Method (ISketchSpline)

Smoothens the shape of a spline that was changed by dragging a node on a [control polygon](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub RelaxSpline()
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
instance.RelaxSpline()
```

C#

```
void RelaxSpline()
```

C++/CLI

```
void RelaxSpline();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::RelaxSpline](#).

Example

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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ResetAllHandles Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : ResetAllHandles Method (ISketchSpline)

Resets all [handles](#) to their initial state.

.NET Syntax

Visual Basic (Declaration)

```
Sub ResetAllHandles()
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
instance.ResetAllHandles()
```

C#

```
void ResetAllHandles()
```

C++/CLI

```
void ResetAllHandles();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Use [SplineHandle::Reset](#) to reset one handle to its initial state.

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::ResetAllHandles](#).

Example

[Set Spline Tangency and Curvature Controls \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetSplineHandleCount Method](#)
[ISketchSpline::GetSplineHandles Method](#)
[ISketchSpline::!GetSplineHandles Method](#)
[ISketchSpline::ShowSplineHandles Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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SetControlVertexWeights Method (ISketchSpline)

[See Also](#)



Sets the weights of the control vetexes of this rational spline.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function SetControlVertexWeights( _  
    ByVal ControlWeightValues As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim ControlWeightValues As System.Object  
Dim value As System.Boolean  
  
value = instance.SetControlVertexWeights(ControlWeightValues)
```

C#

```
System.bool SetControlVertexWeights(  
    System.object ControlWeightValues  
)
```

C++/CLI

```
System.bool SetControlVertexWeights(  
    & System.Object^ ControlWeightValues  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ControlWeightValues
Array of control vertex weights

Return Value

True if weights successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::SetControlVertexWeights](#).

Remarks

This method:

- is valid only if [ISketchSpline::IsRationalCurve](#) is true.
- returns false if the number of weights in ControlWeightValues is not equal to the number of control vertexes of the spline.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetControlVertexWeights Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SetEquationParameters Method (ISketchSpline)

[See Also](#) [Example](#)



Obsolete. Superseded by [ISketchSpline::SetEquationParameters2](#).

Visual Basic (Declaration)

```
Function SetEquationParameters( _
    ByVal YExpression As System.String, _
    ByVal RangeStart As System.Double, _
    ByVal RangeEnd As System.Double, _
    ByVal IsAngleRange As System.Boolean, _
    ByVal RotationAngle As System.Double, _
    ByVal XOffset As System.Double, _
    ByVal YOffset As System.Double, _
    ByVal LockStart As System.Boolean, _
    ByVal LockEnd As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline
Dim YExpression As System.String
Dim RangeStart As System.Double
Dim RangeEnd As System.Double
Dim IsAngleRange As System.Boolean
Dim RotationAngle As System.Double
Dim XOffset As System.Double
Dim YOffset As System.Double
Dim LockStart As System.Boolean
Dim LockEnd As System.Boolean
Dim value As System.Boolean
```

```
value = instance.SetEquationParameters(YExpression, RangeStart, RangeEnd, IsAngleRange,
RotationAngle, XOffset, YOffset, LockStart, LockEnd)
```

C#

```
System.bool SetEquationParameters(
    System.string YExpression,
    System.double RangeStart,
    System.double RangeEnd,
    System.bool IsAngleRange,
    System.double RotationAngle,
    System.double XOffset,
    System.double YOffset,
    System.bool LockStart,
    System.bool LockEnd
)
```

C++/CLI

```
System.bool SetEquationParameters(
    & System.String^ YExpression,
    & System.double RangeStart,
    & System.double RangeEnd,
    & System.bool IsAngleRange,
    & System.double RotationAngle,
    & System.double XOffset,
    & System.double YOffset,
    & System.bool LockStart,
    & System.bool LockEnd
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

YExpression

Equation for y (see **Remarks**)

RangeStart

Start value of x (see **Remarks**)

RangeEnd

End value of x (see **Remarks**)

IsAngleRange

True if the range and x value represent an angle (in radians), false if not

RotationAngle
Rotation angle (in radians) for the curve
XOffset
Offset in x for f(x), where x = 0 (see **Remarks**)
YOffset
Offset in y for f(x), where x = 0 (see **Remarks**)
LockStart
True to lock the start point (RangeStart) of the curve, false to not
LockEnd
True to lock the end point (RangeEnd) of the curve, false to not

Return Value

True if the equation-driven curve's parameters are set, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::SetEquationParameters](#).

Example

[Create Equation-driven Curve \(C#\)](#)
[Create Equation-driven Curve \(VB.NET\)](#)
[Create Equation-driven Curve \(VBA\)](#)

Remarks

YExpression:

Any function that you can define as a 2D function in the following form:

`y-in-the-sketch = f(x-in-the-sketch)`

will appear as a curve in the sketch. For example:

`y = sin(x)`

RangeStart and RangeEnd

You cannot specify string values for RangeStart and RangeEnd; you must specify double values.

XOffset:

In the equation:

`y = sin(x) [x=0 to 2Pi]`

the start point (0,0) is moved by XOffset in the x direction.

YOffset:

In the equation:

`y = sin(x) [x=0 to 2Pi]`

the start point (0,0) is moved by YOffset in the y direction.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchManager::CreateEquationSpline Method](#)
[ISketchSpline::GetEquationParameters Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetEquationParameters2 Method (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : SetEquationParameters2 Method (ISketchSpline)

Sets an equation-driven curve's parameters.

Visual Basic (Declaration)

```
Function SetEquationParameters2( _
    ByVal XExpression As System.String, _
    ByVal YExpression As System.String, _
    ByVal ZExpression As System.String, _
    ByVal RangeStart As System.Double, _
    ByVal RangeEnd As System.Double, _
    ByVal IsAngleRange As System.Boolean, _
    ByVal RotationAngle As System.Double, _
    ByVal XOffset As System.Double, _
    ByVal YOffset As System.Double, _
    ByVal LockStart As System.Boolean, _
    ByVal LockEnd As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline
Dim XExpression As System.String
Dim YExpression As System.String
Dim ZExpression As System.String
Dim RangeStart As System.Double
Dim RangeEnd As System.Double
Dim IsAngleRange As System.Boolean
Dim RotationAngle As System.Double
Dim XOffset As System.Double
Dim YOffset As System.Double
Dim LockStart As System.Boolean
Dim LockEnd As System.Boolean
Dim value As System.Boolean
```

```
value = instance.SetEquationParameters2(XExpression, YExpression, ZExpression, RangeStart,
RangeEnd, IsAngleRange, RotationAngle, XOffset, YOffset, LockStart, LockEnd)
```

C#

```
System.bool SetEquationParameters2(
    System.string XExpression,
    System.string YExpression,
    System.string ZExpression,
    System.double RangeStart,
    System.double RangeEnd,
    System.bool IsAngleRange,
    System.double RotationAngle,
    System.double XOffset,
    System.double YOffset,
    System.bool LockStart,
    System.bool LockEnd
)
```

C++/CLI

```
System.bool SetEquationParameters2(
    & System.String^ XExpression,
    & System.String^ YExpression,
    & System.String^ ZExpression,
    & System.double RangeStart,
    & System.double RangeEnd,
    & System.bool IsAngleRange,
    & System.double RotationAngle,
    & System.double XOffset,
    & System.double YOffset,
    & System.bool LockStart,
    & System.bool LockEnd
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XExpression
Equation for x
YExpression
Equation for y
ZExpression
Equation for z
RangeStart
Start value of x
RangeEnd
End value of x
IsAngleRange
True if the range and x value represent an angle (in radians), false if not
RotationAngle
Rotation angle (in radians) for the curve
XOffset
Offset in x for f(x), where x = 0
YOffset
Offset in y for f(x), where x = 0
LockStart
True to lock the start point (RangeStart) of the curve, false to not
LockEnd
True to lock the end point (RangeEnd) of the curve, false to not

Return Value

True if the equation-driven curve's parameters are set, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::SetEquationParameters2](#).

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetEquationParameters2 Method](#)
[ISketchManager::CreateEquationSpline2 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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ShowCurvatureCombs Property (ISketchSpline)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : ShowCurvatureCombs Property (ISketchSpline)

Gets or sets whether to show curvature combs.

.NET Syntax

Visual Basic (Declaration)

Property ShowCurvatureCombs As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Boolean  
  
instance.ShowCurvatureCombs = value  
  
value = instance.ShowCurvatureCombs
```

C#

```
System.bool ShowCurvatureCombs {get; set;}
```

C++/CLI

```
property System.bool ShowCurvatureCombs {  
    System.bool get();
```

```
void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to show curvature combs, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::ShowCurvatureCombs](#).

Remarks

Use [ISketchManager::CurvatureScale](#) to adjust the size of all of the curvature combs of this spline.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchManager::CurvatureDensity Property](#)
[ISketchSpline::AddCurvatureControl Method](#)
[ISketchHandle::CurvatureControlled Property](#)
[ISketchHandle::RadiusOfCurvature Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

[SOLIDWORKS API Help](#)

[Send comments on this topic.](#)

ShowInflectionPoints Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : ShowInflectionPoints Property (ISketchSpline)

Gets or sets whether show the inflection points of this spline.

.NET Syntax

Visual Basic (Declaration)

```
Property ShowInflectionPoints As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Boolean  
  
instance.ShowInflectionPoints = value  
  
value = instance.ShowInflectionPoints
```

C#

```
System.bool ShowInflectionPoints {get; set;}
```

C++/CLI

```
property System.bool ShowInflectionPoints {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to show the inflection points, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::ShowInflectionPoints](#).

Example

[Get and Set Spline Properties \(VBA\)](#)

Remarks

Inflection points indicate where the concavity of the spline changes.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::DeletePoint Method](#)
[ISketchSpline::GetPointCount Method](#)
[ISketchSpline::GetPoints2 Method](#)
[ISketchSpline::EnumPoints Method](#)
[ISketchSpline::InsertPoint Method](#)
[ISketchHandle::SplinePointNumber Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowMinimumRadius Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : ShowMinimumRadius Property (ISketchSpline)

Gets or sets the minimum radius of a curve for this spline.

.NET Syntax

Visual Basic (Declaration)

Property ShowMinimumRadius As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim value As System.Boolean
```

```
instance.ShowMinimumRadius = value
```

```
value = instance.ShowMinimumRadius
```

C#

```
System.bool ShowMinimumRadius {get; set;}
```

C++/CLI

```
property System.bool ShowMinimumRadius {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Minimum radius

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::ShowMinimumRadius](#).

Example

[Get and Set Spline Properties \(VBA\)](#)

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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ShowSplineHandles Property (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : ShowSplineHandles Property (ISketchSpline)

Gets or sets whether to show the [handles](#) for this spline.

.NET Syntax

Visual Basic (Declaration)

Property ShowSplineHandles As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchSpline
Dim value As System.Boolean

instance.ShowSplineHandles = value

value = instance.ShowSplineHandles
```

C#

```
System.bool ShowSplineHandles {get; set;}
```

C++/CLI

```
property System.bool ShowSplineHandles {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to show spline handles, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::ShowSplineHandles](#).

Example

[Get and Set Spline Properties \(VBA\)](#)

Remarks

If setting this property to True, then use [IModelDoc2::WindowRedraw](#) after to see the spline handles.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)
[ISketchSpline::GetSplineHandleCount Method](#)
[ISketchSpline::GetSplineHandles Method](#)

[ISketchSpline::IGetSplineHandles Method](#)
[ISketchSpline::ResetAllHandles Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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Simplify Method (ISketchSpline)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchSpline Interface](#) : Simplify Method (ISketchSpline)

Reduces the number of points in a spline to increase system performance in models with complex spline curves.

.NET Syntax

Visual Basic (Declaration)

```
Sub Simplify( _  
    ByVal ToleranceIn As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As ISketchSpline  
Dim ToleranceIn As System.Double  
  
instance.Simplify(ToleranceIn)
```

C#

```
void Simplify(  
    System.double ToleranceIn  
)
```

C++/CLI

```
void Simplify(  
& System.double ToleranceIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToleranceIn
Smoothing factor, in meters, to use to simplify sketch

Visual Basic for Applications (VBA) Syntax

See [SketchSpline::Simplify](#).

Example

[Create Equation-driven Curve \(VBA\)](#)
[Create Equation-driven Curve \(VB.NET\)](#)
[Create Equation-driven Curve \(C#\)](#)

Remarks

The sketch must be in edit mode and a spline must be preselected.

See Also

[ISketchSpline Interface](#)
[ISketchSpline Members](#)

Availability

